

Article

An Environmental Impact Assessment of Largemouth Bass (*Micropterus salmoides*) Aquaculture in Hangzhou, China

Haochen Hou ^{1,2}, Anqi Ren ^{1,2}, Lixingbo Yu ^{1,2}, Zhen Ma ^{1,2,*} , Yun Zhang ³ and Ying Liu ^{1,4}

¹ Key Laboratory of Environment Controlled Aquaculture (Dalian Ocean University) Ministry of Education, 52 Heishijiao Street, Dalian 116023, China; houhaochen@dlou.edu.cn (H.H.); raq1054778288@163.com (A.R.); ylixingbo@163.com (L.Y.); liuying@dlou.edu.cn (Y.L.)

² College of Marine Technology and Environment, Dalian Ocean University, 52 Heishijiao Street, Dalian 116023, China

³ Key Laboratory of Industrial Ecology and Environmental Engineering (MOE), School of Environmental Science and Technology, Dalian University of Technology, 2 Linggong Road, Dalian 116024, China; zhangyun@dlut.edu.cn

⁴ College of Biosystems Engineering and Food Science, Zhejiang University, 866 Yuhangtang Road, Hangzhou 310058, China

* Correspondence: mazhen@dlou.edu.cn

Abstract: With the rapid increase in aquaculture production, its role in food safety and nutritional security has become increasingly important, but this has also given rise to environmental problems that cannot be ignored. The largemouth bass (*Micropterus salmoides*) has become a widely cultivated and highly economic freshwater farmed species since it was introduced to China in 1983; however, the environmental impacts of its freshwater pond aquaculture process have not yet been elucidated. Here, life cycle assessment (LCA), a decision-making tool that can evaluate and identify environmental issues during production processes, was used to evaluate the environmental performance of the largemouth bass freshwater pond aquaculture process, and a large-scale, commercial company was selected as an example in Hangzhou, China. The results showed that the pond-farming stage and marine aquatic ecotoxicity potential (MAETP) had the largest environmental impacts on the entire aquaculture process. An environmental contribution analysis indicated that electricity (48%) and emissions (23%) were two key factors in the seed-rearing stage, and electricity (60%) and feeds (26%) were two main impact contributors in the pond-farming stage. Improvement measures based on emerging technologies in aquaculture were discussed, namely, clean energies, industrial pond farming, and intelligent feeding strategies, to help with decision making for continuous improvement in the environmental performance of largemouth bass pond farming. Moreover, suggestions for further aquaculture LCA studies in China were summarized, as they will provide a useful reference for promoting the development of China's aquaculture LCA research and the enrichment of the world's aquaculture life cycle inventory databases.

Keywords: life cycle assessment; energy consumption; environmental impact; industrial pond farming; intelligent feeding strategy



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1. Introduction

Because of the rapid growth in production and innovative techniques, feed ingredients, and supply chain management, the worldwide aquaculture industry is growing up rapidly and has continued to blend into the world food framework. Statistics from the Food and Agriculture Organization of the United Nations (FAO) in 2020 indicated that global aquaculture production had reached 112.6 million tons. In total, 49.2% of the contribution was achieved due to global aquatic animal production [1]. In the future, the aquaculture sector will play an increasingly important role in the sustenance of global food security and supply demand. China is the world's largest aquaculture country; in 2020, China's

aquaculture production was 52.24 million tons [2], accounting for about 46.4% of the world's aquaculture output and playing an important role in almost all aquatic products [3]. This sector also has made important contributions to China's food supply, food security, and its peoples' demand for high-quality aquatic products and protein.

The origin of the largemouth bass (*Micropterus salmoides*) was in California, USA, and it is a popular game fish in North America. It is a type of carnivorous fish with characteristics such as a large food ration, fast growth, low temperature resistance, delicious taste, and rich nutrition [4]. The largemouth bass can survive in a water temperature range of 0–34 °C and start ingestion at above 10 °C, and it has an optimal growth temperature of 20–25 °C. The largemouth bass was introduced to China in 1983, and its artificial seed rearing was conducted in 1985. Nowadays, the largemouth bass has become a freshwater farmed species with a high economic value. In 2021, the total production of largemouth bass in China was 702,093 t [2]. At present, the studies on largemouth bass have been mainly concerned with its feed. Molinari et al. [5] reported the use of live food as a vehicle of soybean meals for the nutritional programming of largemouth bass. Yin et al. [6] evaluated the risks and benefits of dietary consumption for the potential neurodevelopmental effects of largemouth bass aquaculture in China. Moreover, other studies have also focused on the dietary issues [7,8] and environmental factors' effects [9] in its farming process. Currently, the main aquaculture strategy for largemouth bass is pond farming in China. Ponds are the oldest aquaculture production systems, with indications of activity since ancient times in China and Egypt [10]. Over the past three decades, farming in fish ponds has changed from traditional natural farming to intensive farming [11]. However, the rapid development of intensive farming, characterized by high fish-stocking densities and large pellet-feed inputs, has led to serious environmental problems, such as eutrophication and greenhouse gas emissions [12].

Life cycle assessment (LCA), as a decision-making tool to assess the environmental impacts of product life cycle (cradle to grave) processes [13], has become the leading tool for identifying the key environmental impacts of seafood production systems [14]. According to a review of the Web of Science core collection database, with literature publication dates selected from January 2000 to December 2022, approximately 80 research papers on fish aquaculture LCA have been published. The earliest study on fish aquaculture LCA was a feed production environmental impact assessment of French rainbow trout (*Oncorhynchus mykiss*) conducted by Papatryphon et al. [15]. Subsequently, researchers have focused on single-species farming systems, for instance, the Finnish rainbow trout system [16] and global Atlantic salmon (*Salmo salar*) system [17]. As there has been an improvement in the intensive level and system complexity of global fish farming, LCAs of fish farming have begun to expand to broader system boundaries; Aubin et al. [18] compared the life cycle environmental impacts of Atlantic-salmon-farming systems in Norway, UK, Canada, and Chile and pointed out that the emissions per unit production were significantly different in different regions. In addition, Aubin et al. [19] conducted a comparative life cycle analysis of carnivorous finned fish farming systems in different regions, including the freshwater-rainbow-trout-farming system in France, European seabass (*Dicentrarchus labrax*) deep-sea cage farming system in Greece, and turbot (*Scophthalmus maximus*) land-based recirculating aquaculture system in France. Chen et al. [20] used rainbow trout farming in the Heilongjiang Province and Beijing City in China as an example to perform comparative LCAs, and the results indicated the environmental impact contributions at each stage. Recently, the large-scale production of many emerging fish species has been gradually achieved on the basis of continuous innovation and the development of fish-farming technologies. Researchers have published LCA studies on catfish (*Pangasianodon hypophthalmus*) [21,22], carp (*Cyprinus carpio*) [23], and tiger puffer (*Takifugu rubripes*) [24]. Because Atlantic salmon is one of the most economically valuable farmed species worldwide, most fish LCA studies have focused on this species [17,25,26]. Gephart et al. [27] found that the farming of silver carp (*Hypophthalmichthys molitrix*) and bighead carp (*Aristichthys nobilis*) emitted the lowest levels of greenhouse gases, nitrogen,

and phosphorus, and the highest levels of corresponding water consumption. In contrast, farming salmon and trout can reduce the use of land and water resources. Bohnes et al. [28] appealed to Asia for more studies on aquaculture LCA. In recent years, many studies have used LCAs to survey the quantitative results of environmental problems in pond farming and certified that the method is very suitable for this research. Biermann and Geist [29] applied an LCA to compare the environmental impacts of conventional and organic carp (*Cyprinus carpio* L.) reared in conventional pond aquaculture, and they showed that feed and pond dredging were the main contributors to the environmental impacts. Fonseca et al. [30] reported that feed, services, and water were the main inputs to the rural freshwater pond aquaculture systems of the yellow-tail lambari (*Astyanax lacustris*). Pelletier and Tydmers [31] concluded that the application of an LCA plays a significant role in the environmental improvement of the tilapia (*Oreochromis niloticus*)-intensive aquaculture systems in lakes and ponds in Indonesia. These studies analyzed the environmental impacts of each kind of farmed fish species and offered improvement measures and decision-making suggestions, which will greatly increase the number of aquaculture LCA studies and provide a reference for the quantitative assessment and continuous improvement of fish aquaculture environmental performance.

Do the eutrophication and greenhouse gas emissions problems also appear during the largemouth bass freshwater pond aquaculture process? What are the key factors that affect the environmental impact results? Which measures can improve the environmental performance of largemouth bass freshwater pond aquaculture in China? These issues have not yet been resolved. Moreover, no studies have reported the environmental impacts of largemouth bass farming on the basis of an LCA. Therefore, in this study, the LCA method was used to conduct an environmental impact assessment of largemouth bass freshwater pond aquaculture, and the key factors that affect the environmental impacts of the largemouth bass farming process were systematically identified to help farming enterprises and governments to determine the key issues of environmental performance and provide improvement measures. The purpose of this study was to enrich the world's aquaculture basic life cycle inventory (LCI) databases and provide technical support and practical guidance for LCAs of other fish pond aquaculture processes in China.

2. Materials and Methods

Two basic international standards [32,33] were executed and the LCA method was employed to assess the environmental impact factors in largemouth bass freshwater pond aquaculture.

2.1. Goal and Scope Definition

Zhejiang Hengze Ecological Agriculture Technology Co., Ltd., a large-scale, commercial largemouth bass pond cultivation company in Hangzhou, China, was selected. It is a famous land-based aquaculture company in China and it has won the title of National Modern Agricultural Science and Technology Demonstration Base, the title of Agricultural Science and Technology Enterprise in Zhejiang, and the title of Healthy Aquaculture Demonstration Farm of the Ministry of Agriculture and Rural Affairs of China. The case company has 4.5 million USD of registered capital, and its principal aquatic production is largemouth bass. The company has ponds that cover an area of 953,000 m². The production situation of the enterprise can represent an advanced level of largemouth bass aquaculture in China. Largemouth bass with a live weight of 1 t, harvested at the pond-farming stage, were used as the functional units.

2.1.1. The Stage Description of Seed Rearing

Before the seed-rearing stage, a sanitizer (quicklime) was used to disinfect the farming tank. Translucent largemouth bass larvae (about 0.7 cm in length) were cultured in the recirculating aquaculture workshop. The feed was brine shrimp (*Artemia sinica*). When the larvae grew to 3–4 cm, they were fed with pellet feed. When the larvae grew to 7 cm, they

were transported to a land-based farming pond via a live fish track. The whole breeding process lasted for 60 days. The main resources and energy inputs during the rearing stage were disinfectants, fresh water, electricity, feed, and gasoline. The main air pollutants were carbon dioxide (CO_2), NO_x , and sulphur dioxide (SO_2). The major wastewater pollutants were chemical oxygen demand (COD), total phosphorus (P), and total nitrogen (N).

2.1.2. The Stage Description of Pond Farming

For the pond-farming stage, the fish ponds were disinfected with quicklime. During the pond farming, fresh water was pumped from the ground to the pond every day to maintain the water quality, and the largemouth bass were fed with pellet feed. Aerators were used to maintain the dissolved oxygen at more than 4.5 mg/L, and the opening time was 10 h every day. After 150 days, when the live weight of a single fish reached 500 g, the bass were caught to sell. The total N, total P, and COD were also the main emissions in the wastewater and feed, electricity, and freshwater were used in this stage.

The schematic diagram for largemouth bass freshwater pond aquaculture is revealed in Figure 1.

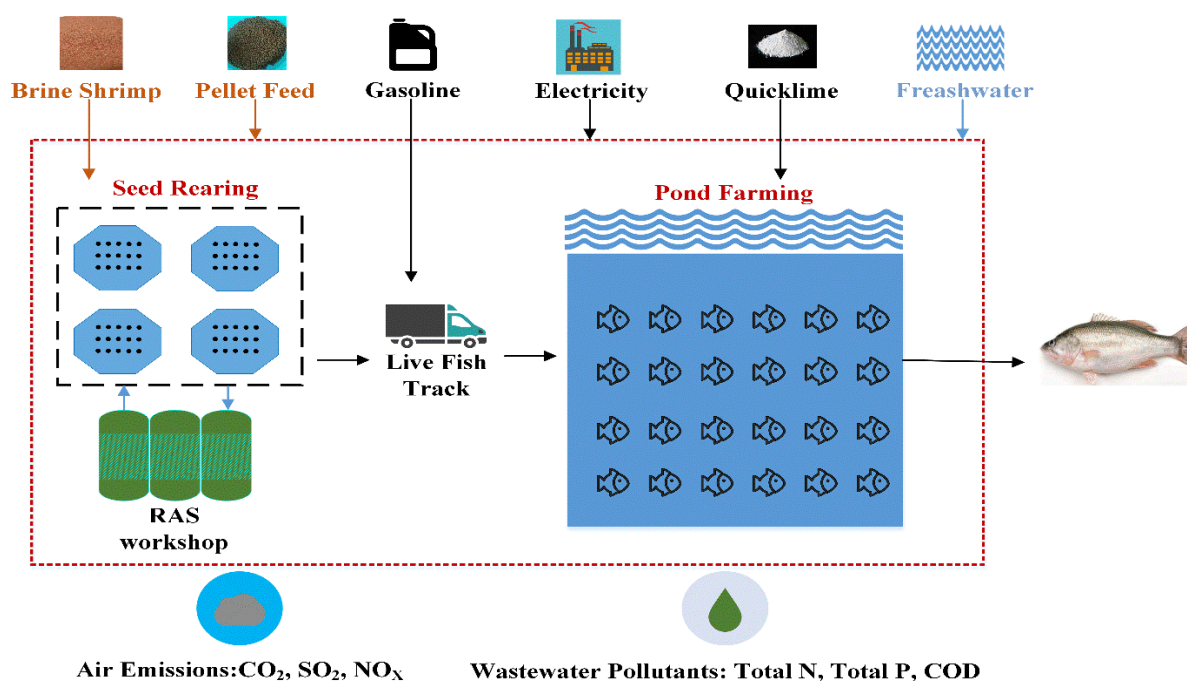


Figure 1. Schematic diagram of the largemouth bass freshwater pond aquaculture process.

2.2. Inventory Analysis

Table 1 provides the data of the largemouth bass pond aquaculture process. The electricity, feed, gasoline, sanitizer (quick lime), and freshwater were obtained from the actual data of the case company. The concentrations of the total N, total P, and COD in the wastewater were monitored in the lab. The GaBi Professional database 2021 offered the data for the energy and material inputs; the Chinese Life Cycle Database (CLCD, Eke Co., Ltd.) supplied the pellet feed production process data. The Ecoinvent 3.7 database calculated the other air emission data.

Table 1. Life cycle inventory for two stages in the largemouth bass freshwater pond aquaculture process.

	Object	Units	Seed Rearing	Pond Farming
Input	Electricity	kWh	61.15	2190
	Pellet feed	kg	19.25	1100
	Quick lime	kg	4.75	42.85
	Gasoline	L	16.50	280
	Freshwater	m ³	37.13	1350
Output	Brine Shrimp	kg	12.84	--
	Carbon Dioxide	kg	37.95	644
	Sulphur Dioxide	kg	0.14	2.38
	NO _x	kg	0.12	2.07
	Total N	kg	0.17	6.65
	Total P	kg	0.028	1.35
	COD	kg	0.91	15.75

2.3. Life Cycle Impact Assessment

An impact assessment is an important part of a life cycle assessment. The optional assessment methods include weighting, normalization, and characterization. The formulas for the characterization (Equation (1)) and normalization (Equation (2)) calculation steps of the life cycle assessment can be formulated as:

$$\text{Characterization results} = \sum_i m_i \times \text{Characterization factor} \quad (1)$$

where m_i represents the quantification results of the input or output of the i th substance within the system boundary (e.g., pollutant emissions, resource and energy consumption, resource and energy exploitation, and land use, etc.)

$$\text{Normalization results} = \frac{\text{Characterization results}}{\text{Normalization reference value}} \quad (2)$$

In the present research, process data were used to analyze the environmental performance of largemouth bass freshwater pond aquaculture. A problem-oriented approach CML-IA-Aug. 2016-world method was used and an LCA using the Experts 10.5 software (academy version) was employed to calculate the LCA results. The impact categories used in the assessment process are shown in Table 2.

Table 2. Eleven LCA impact categories in CML-IA-Aug. 2016-world method.

Category	Abbreviation	Characterization Units
Abiotic Depletion Potential (elements)	ADPe	kg Sb eq.
Abiotic Depletion Potential (fossil)	ADPf	MJ
Acidification Potential	AP	kg SO ₂ eq.
Eutrophication Potential	EP	kg Phosphate eq.
Freshwater Aquatic Ecotoxicity Potential	FAETP	kg DCB eq.
Global Warming Potential	GWP	kg CO ₂ eq.
Human Toxicity Potential	HTP	kg DCB eq.
Marine Aquatic Ecotoxicity Potential	MAETP	kg DCB eq.
Ozone Layer Depletion Potential	ODP	kg R11 eq.
Photochemical Ozone Creation Potential	POCP	kg Ethene eq.
Terrestrial Ecotoxicity Potential	TEP	kg DCB eq.

3. Results

3.1. Characterization Results

The characterization results of the largemouth bass freshwater pond aquaculture process are provided in Table 3 and the contributions in each category in the two stages are shown in Figure 2.

Table 3. Life cycle characterization results for two stages in the largemouth bass freshwater pond aquaculture process.

Category/Stage	Units	Seed Rearing	Pond Farming
ADPe	kg Sb eq.	7.96×10^{-6}	2.24×10^{-4}
ADPf	MJ	1.47×10^3	4.38×10^4
AP	kg SO ₂ eq.	0.50	17.90
EP	kg Phosphate eq.	1.91	8.93
FAETP	kg DCB eq.	0.56	12.90
GWP	kg CO ₂ eq.	1.24×10^2	4.44×10^3
HTP	kg DCB eq.	6.58	2.50×10^2
MAETP	kg DCB eq.	6.72×10^3	2.62×10^5
ODP	kg R11 eq.	2.83×10^{-10}	1.28×10^{-7}
POCP	kg Ethene eq.	3.78×10^{-2}	1.41
TEP	kg DCB eq.	8.55×10^{-2}	2.92

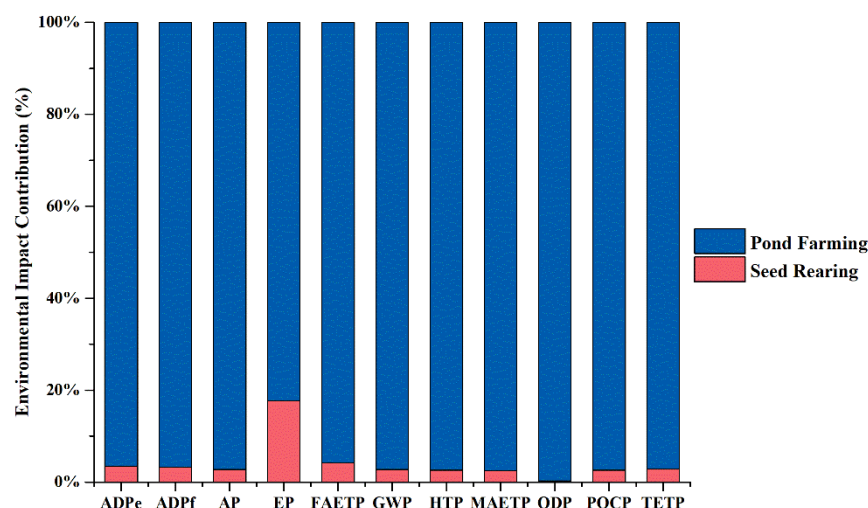


Figure 2. Environmental contribution analysis of the characterization results of the largemouth bass freshwater pond aquaculture process.

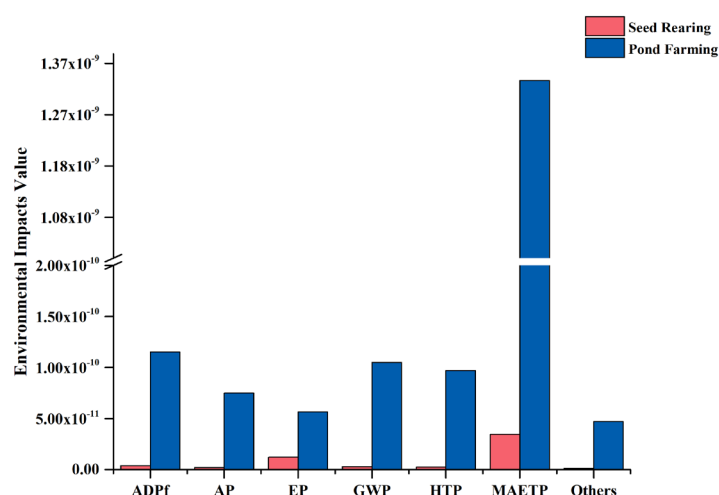
The contribution of the pond-farming stage to all 11 environmental impact categories was the greatest, and it is obvious that this stage is critical in the entire largemouth bass pond-farming process (Figure 2). The category with the highest percentage was ODP (99.78% with 1.28×10^{-7} kg R11 eq.), and the category with the lowest percentage was EP (82.33% with 8.93 kg phosphate eq.).

3.2. Normalization Results

In order to make a more detailed comparative analysis of the environmental impact of each category in the largemouth bass freshwater pond aquaculture process and find out the key impact factors and opportunities to prevent pollution, the normalization results were calculated and are shown in Table 4. On the basis of the results, the ADP (fossil), AP, EP, GWP, HTP, and MAETP impact categories were selected, as they showed larger environmental impact potential than the other categories and were also related to energy consumption, eco-systems, and human health; the other categories were combined as “others” (Figure 3).

Table 4. Life cycle normalization results for the two stages in the largemouth bass freshwater pond aquaculture process.

Category	Units	Seed Rearing	Pond Farming
ADPe	yr	2.21×10^{-14}	6.23×10^{-13}
ADPf	yr	3.86×10^{-12}	1.15×10^{-10}
AP	yr	2.09×10^{-12}	7.49×10^{-11}
EP	yr	1.22×10^{-11}	5.65×10^{-11}
FAETP	yr	2.37×10^{-13}	5.46×10^{-12}
GWP	yr	2.93×10^{-12}	1.05×10^{-10}
HTP	yr	2.55×10^{-12}	9.70×10^{-11}
MAETP	yr	3.45×10^{-11}	1.34×10^{-9}
ODP	yr	1.25×10^{-18}	5.60×10^{-16}
POCP	yr	1.03×10^{-12}	3.84×10^{-11}
TEP	yr	7.85×10^{-14}	2.68×10^{-12}
TOTAL	yr	5.95×10^{-11}	1.84×10^{-9}

**Figure 3.** Environmental impact potential analysis of the two stages in the largemouth bass freshwater pond aquaculture process.

On the basis of the comparative analysis of the normalization results for the environmental impact in each stage and category, MAETP had the largest environmental contribution in the seed-rearing and pond-farming stages (3.45×10^{-11} yr and 1.34×10^{-9} yr, respectively). The main reason for this is that a large amount of pellet feed needed to be invested in the pond-farming stage to maintain the growth of the largemouth bass and energy consumption during the feed production process, making it the key impact factor. Moreover, electricity was mainly consumed to provide power for the farming machines to maintain the water quality and feeding work (pumps, aerators, feeder, and other equipment), making the categories of ADPf and GWP contribute more environmental impacts in the two stages. The concentrations of the total N, total P, and COD were 4.84 mg/L, 0.75 mg/L, and 24.51 mg/L, respectively, in the seed-rearing stage, and were 4.92 mg/L, 0.98 mg/L, and 11.66 mg/L, respectively, in the pond-farming stage. The pollutants from the case company just met the secondary standards for water discharge from freshwater aquaculture ponds regulated in China (total N ≤ 5 mg/L, total P ≤ 1 mg/L, and COD ≤ 25 mg/L). Thus, the pollutants discharged in the wastewater caused AP, HTP, and EP to become the major environmental impact categories in the farming process.

3.3. Monte Carlo Simulation Results

Monte Carlo simulation is a helpful method for revealing the influence of uncertainty [34], and it has been widely used in the uncertainty research of LCAs. In this study, a

Monte Carlo simulation was performed with 1000 such rankings, and the 95% confidence interval parameter was set. The simulation results showed that the trends of the uncertainty ranges of each category did not change significantly (Table 5).

Table 5. Monte Carlo simulation results in largemouth bass freshwater pond aquaculture process.

Stages	Impact Category	Normalization Results (yr)	Monte Carlo Simulation Results		
			Confidence Interval 95% (yr)	Mean (yr)	SD (yr)
Seed Rearing	ADPe	2.21×10^{-14}	2.18×10^{-14} – 2.24×10^{-14}	2.20×10^{-14}	2.54×10^{-14}
	ADPf	3.86×10^{-12}	3.81×10^{-12} – 3.92×10^{-12}	3.87×10^{-12}	3.95×10^{-14}
	AP	2.09×10^{-12}	2.06×10^{-12} – 2.12×10^{-12}	2.08×10^{-12}	2.10×10^{-14}
	EP	1.22×10^{-11}	1.20×10^{-11} – 1.24×10^{-11}	1.21×10^{-11}	1.27×10^{-13}
	FAETP	2.37×10^{-13}	2.34×10^{-13} – 2.40×10^{-13}	2.36×10^{-13}	2.47×10^{-15}
	GWP	2.93×10^{-12}	2.89×10^{-12} – 2.97×10^{-12}	2.94×10^{-12}	2.95×10^{-14}
	HTP	2.55×10^{-12}	2.52×10^{-12} – 2.59×10^{-12}	2.57×10^{-12}	2.54×10^{-14}
	MAETP	3.45×10^{-11}	3.40×10^{-11} – 3.50×10^{-11}	3.46×10^{-11}	3.63×10^{-13}
	ODP	1.25×10^{-18}	1.23×10^{-18} – 1.27×10^{-18}	1.26×10^{-18}	1.30×10^{-20}
	POCP	1.03×10^{-12}	1.02×10^{-12} – 1.04×10^{-12}	1.04×10^{-12}	1.07×10^{-14}
Pond Farming	TEP	7.85×10^{-14}	7.74×10^{-14} – 7.96×10^{-14}	7.86×10^{-14}	8.09×10^{-16}
	ADPe	6.23×10^{-13}	6.14×10^{-13} – 6.31×10^{-13}	6.24×10^{-13}	6.34×10^{-15}
	ADPf	1.15×10^{-10}	1.13×10^{-10} – 1.17×10^{-10}	1.14×10^{-10}	1.19×10^{-12}
	AP	7.49×10^{-11}	7.38×10^{-11} – 7.59×10^{-11}	7.48×10^{-11}	7.68×10^{-13}
	EP	5.65×10^{-11}	5.57×10^{-11} – 5.73×10^{-11}	5.66×10^{-11}	5.72×10^{-13}
	FAETP	5.46×10^{-12}	5.38×10^{-12} – 5.54×10^{-12}	5.47×10^{-12}	5.78×10^{-14}
	GWP	1.05×10^{-10}	1.04×10^{-10} – 1.06×10^{-10}	1.04×10^{-10}	1.02×10^{-12}
	HTP	9.70×10^{-11}	9.56×10^{-11} – 9.83×10^{-11}	9.71×10^{-11}	1.00×10^{-12}
	MAETP	1.34×10^{-9}	1.32×10^{-9} – 1.36×10^{-9}	1.32×10^{-9}	1.36×10^{-11}
	ODP	5.60×10^{-16}	5.51×10^{-16} – 5.68×10^{-16}	5.62×10^{-16}	5.94×10^{-18}
	POCP	3.84×10^{-11}	3.78×10^{-11} – 3.89×10^{-11}	3.83×10^{-11}	4.02×10^{-13}
	TEP	2.68×10^{-12}	2.64×10^{-12} – 2.72×10^{-12}	2.67×10^{-12}	2.74×10^{-14}

4. Discussion

4.1. Environmental Contribution Analysis

According to the LCA results in this study, the pond-farming stage and MAETP showed the largest environmental impact contributions for the entire largemouth bass freshwater pond aquaculture process. Moreover, the environmental contributions of electricity, gasoline, emissions, and feeds were classified. Electricity (48%) and emissions (23%) were two key factors in the seed-rearing stage, and electricity (60%) and feeds (26%) were two major impact contributors in the pond-farming stage (Figure 4). Feed production and electricity usage were still key issues that affected the fish pond aquaculture process [29–31]. Compared to open-water aquaculture systems [22,35], the environmental impact of electricity consumption was greater than that of feed production, the main reason being that electricity was mainly consumed to provide power for the farming machines to maintain the water quality and feeding work, but electricity in open-water aquaculture systems is only used in feed production, the main type of energy consumption is gasoline, and the environmental impacts are much lower than feed production. These results came to the same conclusions as those in previous LCA studies on China's aquaculture [9,24], and these issues will continue to be major environmental problems for China's aquaculture industry in the next few years.

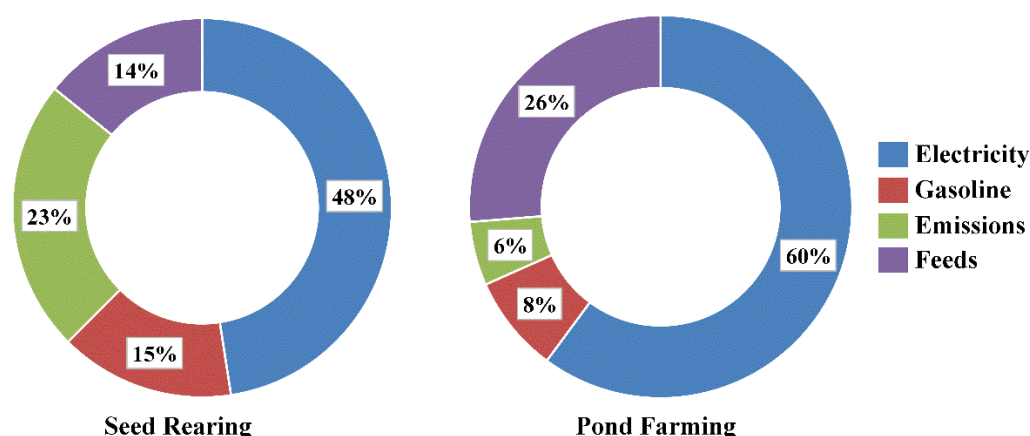


Figure 4. Environmental impact contributions of electricity, gasoline, emissions, and feeds.

In this study, electricity was the main energy type to provide power supply for the aerators, feeders, and water pumps, which caused major environmental stress. Hou et al. [24] discussed the advantages of clean energy environmental performance in aquaculture, and future environmental impact improvement measures are focused on a shift in energy type, using clean energy sources such as solar and wind, rather than traditional coal power. This measure will also be a key part of China's carbon neutral strategy. Moreover, the environmental impact of feed production on the supply chain has to be reduced; more environmentally friendly feeds should be used [36] to replace the pellet feed made with crops, in order to effectively reduce the environmental problems caused by feed production and consumption under the premise of providing the same nutrients.

4.2. Industrial Pond Farming

The industrial pond farming mode was transformed based on the upgrade of traditional pond farming and it can comprehensively coordinate economic and ecological benefits. It has become a strategy for the sustainable development of fish ponds, and it has been widely accepted by China's government, farmers, and markets. Currently, integrated multi-trophic aquaculture, farming efficiency analyses, and wastewater treatment are the main research hotspots. Zhang et al. [37] reported the co-culture mode of largemouth bass and white shrimp (*Litopenaeus vannamei*) in fish ponds, and the economic benefits were 1.61 USD/m². On the basis of the topography and economic situation of the coastal areas in northern China, a new industrial mode that combines greenhouses and deep-well seawater has been built to reduce the environmental impacts of turbot (*Scophthalmus maximus*) pond farming. These measures may be useful strategies for increasing the economic benefits of the freshwater pond aquaculture process for largemouth bass.

On the basis of the LCI in this study, the pollutants (total N, total P, and COD) from the case company just met the secondary standards for water discharge from freshwater aquaculture ponds regulated in China. Currently, pond wastewater is mostly discharged directly into rivers in China, and a national strategy to build a pond wastewater treatment technical system is currently being discussed. Sidoruk and Cymes [38] evaluated three water management systems commonly used in rainbow trout farming, suggesting that the water discharged from fish ponds can have an impact on the water quality of the receiving water bodies and that appropriate technical measures should be taken to reduce the risk of water pollution. The physical methods used in wastewater treatment mainly include precipitation, filtration, foam separation, magnetic separation, and ultraviolet sterilization. Moreover, a certain amount of microalgae can be cultivated in aquaculture wastewater or fish ponds, which absorb the excess nutrients. Jung et al. [39] found that tilapia ponds cultured with *Scenedesmus* and *Chlorella vulgaris* reduced their water exchange by 82% compared to the control group. Therefore, 5% of the total environmental impacts can be reduced when this technology is adopted in largemouth bass pond aquaculture and

the pollution in wastewater will reach the first standards for the water discharge from freshwater aquaculture ponds regulated in China (total N $\leq$ 3 mg/L, total P $\leq$ 0.5 mg/L, and COD $\leq$ 15 mg/L). Ajala and Alexander [40] also demonstrated that *Oocystis minuta*, *Scenedesmus obliquus*, and *Chlorella vulgaris* effectively remove sulfate, nitrate, and phosphate enrichment from wastewater. Therefore, it is quite necessary to establish wastewater treatment systems based on microalgal bioreactor technology or use physical methods to remove pollutants before wastewater is discharged into rivers during the freshwater pond aquaculture process for largemouth bass.

4.3. Intelligent Feeding Strategies

The main factors that determine the cost of aquaculture production and water quality are feeding. Feed production and consumption were key issues that affected the environmental performance of the largemouth bass farming. It is critical for fish culture to offer a nutritionally balanced diet in sufficient amounts at the right times [41]. Fish feeding patterns are extremely diverse, which means there is no single answer to the question of when and how much to feed. With the gradual penetration of modern information technology into various fields of agriculture, research on the intelligent management of aquaculture based on fish behavior is booming [42]. Fish feeding is generally performed by automatic feeders, but this may lead to overfeeding or underfeeding. Effective identifications of the feeding behavior of fish provide an optimal feeding basis, which can reduce resource waste and improve growth rates. For example, the temporal and spatial indexes around the feeding time are significantly different from those of other behaviors. For intelligent feeding strategies, it is important to understand how fish feel and why they behave the way they do, because fish interact with and adapt to their environment through their behavior, creating a link between physiological and ecological events [43]. In addition, fish behavior monitoring can provide the information required to guide disease diagnoses and environmental management in aquaculture.

Recently, many intelligent feeding control methods have been developed, such as computer vision, acoustic methods, and mathematical models [44]. For instance, an adaptive neuro-fuzzy inference system for grass carp (*Ctenopharyngodon idellus*) feed decision making was proposed to improve their feeding efficiency [45]. Zhou et al. [46] published a machine vision and convolutional neural network method with a classification accuracy of 90% to evaluate the feeding intensity of aquaculture fish, which could reduce the feed amount by 10–15%. This innovative method may decrease the total environmental impacts in the largemouth bass aquaculture process by 20%. Moreover, a support vector machine, artificial neural networks, and multiple linear regression were used to develop an intelligent feeding technique in a recirculating aquaculture system for rearing white shrimp [47]. Computer vision technology has assisted intelligent feeding in different ways such as underwater image preprocessing, fish weight and length detection, fish behavior analyses, fish target detection, and intelligent fish-feeding decisions. An et al. [48] proposed that a combination of intelligent feeding and computer vision will contribute to increases in aquaculture production. Similarly, deep learning has been found to create both new opportunities and a series of challenges for information and data processing in smart fish farming [49]. Thus, modern technologies, such as artificial intelligence, big data, the internet of things, 5G, machine vision, and robots [42], will be gradually incorporated into largemouth bass pond farming and the aquaculture industry in the future.

4.4. Suggestions for Further Aquaculture LCA Studies in China

Nowadays, aquaculture LCA studies mainly rely on interviews of enterprises and farms to obtain data in China. Because of limitations in the relationship between LCA researchers and enterprises, most of the actual data sources are obtained from cities or provinces where the aquaculture LCA researchers are located. Therefore, China's government may promote the disclosure of national aquaculture operation data and establish an information-sharing database to enable aquaculture LCA researchers to obtain such data

rapidly and accurately, and provide useful environmental impact improvement measures and recommendations for China's aquaculture industry.

China is the largest aquaculture output country in the world, but the aquaculture LCA research in China is relatively less than that in European countries. According to the current situation of aquaculture in China, many species such as carp, bivalves, and algae can be researched to analyze the environmental impacts of their aquaculture by using the LCA method. In addition, aquaculture has obvious regional characteristics, and the environmental impact of the same species in different countries of the world or different regions in China may be the same or significantly different. Therefore, it is necessary to conduct comparative LCA studies of the same species in different locations in the future to make environmental impact improvement measures more applicable.

5. Conclusions

As far as we know, this is the first LCA study to evaluate the environmental impacts of the largemouth bass freshwater pond aquaculture process. According to the LCA results in this study, the contributions of the environmental impacts in the pond-farming stage and MAETP were the largest for the entire largemouth bass freshwater pond aquaculture process (3.45×10^{-11} in seed rearing and 1.34×10^{-9} in pond farming, respectively), and the environmental contribution analysis indicated that electricity (48%) and emissions (23%) were two key factors in the seed-rearing stage, and electricity (60%) and feeds (26%) were two main impact contributors in the pond-farming stage. These conclusions are consistent with those of previous aquaculture LCA studies, and these issues will continue to be major environmental problems faced by China's aquaculture industry in the next few years. The improvement measures proposed in this study focused on new farming modes (industrial pond farming) and emerging technologies (intelligent feeding strategies).

Moreover, on the basis of previous aquaculture LCA research, suggestions for further studies in China were summarized. The disclosure of national aquaculture operation data and the establishment of an information-sharing database are needed, and more aquaculture LCA studies for China's characteristic species and comparative LCA studies for the same species in different locations are suggested to evaluate the environmental impacts on China's aquaculture and contribution to the world's LCI databases.

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